HAVING LOCALLY NOETHERIAN FIBRES IS FPPF LOCAL ON SOURCE AND TARGET

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5577-5581 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5583-5604, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Locally Noetherian schemes, fibres of a morphism, and fppf descent; the reductions are retained.

Source reference. Original lemma 034C: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5604–5604. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 038R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5604–5604. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The property $\mathcal{P}(f) =$"the fibres of f are locally Noetherian" is local in the fppf topology on the source and the target.*

Proof. Let $f : X \to Y$ be a morphism of schemes. Let $\{\varphi_i : Y_i \to Y\}_{i \in I}$ be an fppf covering of Y. Denote $f_i : X_i \to Y_i$ the base change of f by φ_i. Let $i \in I$ and let $y_i \in Y_i$ be a point. Set $y = \varphi_i(y_i)$. Note that

$$X_{i,y_i} = \mathrm{Spec}(\kappa(y_i)) \times_{\mathrm{Spec}(\kappa(y))} X_y.$$

Moreover, as φ_i is of finite presentation the field extension $\kappa(y_i)/\kappa(y)$ is finitely generated. Hence in this situation we have that X_y is locally Noetherian if and only if X_{i,y_i} is locally Noetherian, see Varieties, Lemma 038R. This fact implies locality on the target.

Let $\{X_i \to X\}$ be an fppf covering of X. Let $y \in Y$. In this case $\{X_{i,y} \to X_y\}$ is an fppf covering of the fibre. Hence the locality on the source follows from Descent, Lemma 034C. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.